Progressive exercises

E1 · Direct application. A code is a sequence of four digits chosen from 0, 1, . . . , 9. Repetitions and the number 0 in the first position are allowed.
minus a 0?

E2 · Application and justification. Ten athletes take part in a race without ex æquo.
Different podiums, with a first, a second and a third place, can we get?

E3 · Method to be organised. Among seven distinct persons, a group of three persons is formed.
How many groups can be formed?

E4 · Choice of strategy. On a grid, we're going from $(0;0)$ to achieve $(4;3)$ by performing exactly four moves from one unit to the right and three to the top. How many paths
Can we follow each other?

E5 · Synthesis or demonstration. There are eight girls and seven boys in one class.
Four students with exactly two girls can be trained?